

Table 1. Notation guide for the main quantities used in Section 3.

Symbol / Term	Meaning
T	Number of tokens in the generated text.
T_s	Number of sentences in the generated text.
$p_t(\cdot)$	Baseline conditional distribution at step t .
$q_t(\cdot)$	Watermarked conditional distribution at step t .
P^s	Sequence distribution induced by the baseline sampler.
Q	Sequence distribution induced by the watermarked sampler.
δ	Bias strength in biased token-level sampling.
g_t	Baseline green-set mass at step t .
$\hat{\sigma}^2$	Variance term for bias-free token-level watermarking.
Z_t	Semantic evidence bit at sentence step t .
ρ	Semantic bias parameter controlling watermark strength.
D_0	Per-unit information budget before edits.
$\text{TV}(P^s, Q)$	Total variation between the baseline and watermarked distributions.
$\text{KL}(Q \ P^s)$	KL divergence from the watermarked distribution to the baseline distribution.
$\text{Detect}_{\text{IT}}$	Information-theoretic detectability.
$\text{Detect}_{\text{comp}}$	Computational detectability over efficient detectors.
keyless observer	Detector that only observes generated text, without access to the key, model probabilities, or watermark rule.
D_λ	A randomized detector operating at security / problem size λ .
$\text{Adv}_{D_\lambda}(\lambda)$	Distinguishing advantage of detector D_λ , i.e., how well it separates watermarked from unwatermarked text.
$\text{ED}(y, \tilde{y})$	Edit distance between original text y and edited text $\tilde{y}$.
ε	Token edit rate after post-generation editing.
ε_s	Probability that editing flips the semantic evidence bit.
$C_{\text{tok}}(\varepsilon)$	Usable post-edit information budget for token-level watermarking.
$C_{\text{sem}}(\varepsilon)$	Usable post-edit information budget for semantic watermarking.
α	False-alarm level of the detector.
β	Target miss probability of the detector.
$D_{\text{req}}^{\text{tok}}$	Required information budget for token-level schemes under the given edit regime.
$D_{\text{req}}^{\text{sem}}$	Required information budget for semantic schemes under the given edit regime.
ℓ	Average number of tokens per sentence.